Review diagnostics and follow-up design

AgencyBuy research record

11 November 2026

Assessment

The review identifies substantive gaps. The existing evidence does not establish a general accuracy improvement from context expansion. The three implementations should not be presented as one validated method. Native-prompt-capped completion of complete source segments is the candidate for the next methods study; the original body-budget and character-trimmed variants remain historical diagnostics.

This report adds no reader inference. It audits all 3,897 internal held-out questions and 480 development questions using archived source contexts, selected spans, benchmark support annotations and model outcomes. The analyses below are post hoc. They do not create a new held-out confirmation, select a favorable budget, or replace the original paired tests.

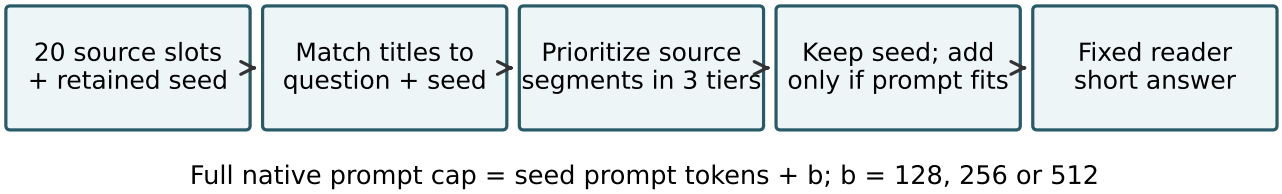


Figure 1: Candidate operator: title matching, source priority and a full-prompt token cap.

Seed and whole-context evidence

The following table gives equal weight to HotpotQA and 2Wiki. The matched rows show Qwen's native-length controls. The original seed, completion and EXIT contexts are shared between readers. Gold title presence is an article-level coverage proxy, not exact gold paragraph retrieval recall. Gold sentence coverage requires a complete normalized benchmark sentence to occur contiguously within one rendered passage with the corresponding title. It does not establish semantic entailment; differences between benchmark contexts and the 2018 DPR corpus can lower this measure. Title prefixes can also interrupt literal sentence matches without deleting their underlying facts. Boolean yes/no answers are excluded from answer-string containment.

Context	Gold title presence (%)	Gold sentence match (%)	Answer in body (%)	Available fact retention (%)
exit	59.04	36.60	58.35	82.78
linked	59.58	32.43	55.05	74.72
linked_equal	59.39	32.39	55.05	74.59
seed	45.21	22.41	44.51	51.08
window_equal	54.39	33.84	55.70	78.10

Context	Gold title presence (%)	Gold sentence match (%)	Answer in body (%)	Available fact retention (%)
raw20	63.56	43.35	70.82	100.00

All support indices for the 4,377 evaluated questions resolved to actual benchmark sentences. Some annotations elsewhere in the full HotpotQA validation source have out-of-range indices; those unrelated questions were not silently used in this evaluation. Conditional retention uses only support sentences that were lexically present in the raw source slots; within each dataset it is a micro ratio, then the datasets are weighted equally.

The seed loses a substantial amount of source-available evidence. Completion recovers part of it. However, matched window has greater literal support coverage and answer presence than matched completion. Evidence retention alone therefore does not explain a completion advantage. This also does not directly measure antecedent resolution or connecting evidence: that stronger mechanism claim needs a dedicated annotation or intervention.

Development scores including b=0

Reader	b	Seed EM	Window EM	Completion EM	Seed F1	Window F1	Completion F1
qwen7	0	35.21	nan	nan	45.56	nan	nan
qwen7	128	35.21	36.25	35.62	45.56	46.85	46.04
qwen7	256	35.21	35.21	38.33	45.56	46.62	49.13
qwen7	512	35.21	36.67	36.88	45.56	48.13	48.01
llama8	0	35.42	nan	nan	45.68	nan	nan
llama8	128	35.42	37.29	36.25	45.68	47.82	47.27
llama8	256	35.42	37.50	35.83	45.68	48.03	46.35
llama8	512	35.42	35.62	39.58	45.68	46.16	50.18

Seed values are from the earlier reader execution profile. Native-cap window/completion values are from the later sweep, on the same questions. This table supplies the missing context; it is not a new same-execution paired test against seed. Fresh seed outputs are included in the fixed follow-up matrix. Every budget remains reported.

Header costs and source duplication

Define header contribution as the token count of the actual full prompt minus that of the same prompt after removing every identifier/title line. Define body contribution as the header-free prompt count minus the empty-evidence prompt count. The two contributions and fixed prompt tokens exactly reconstruct every measured prompt, including tokenizer boundary effects. They are not standalone additive estimates of the title strings.

view	reader	total_added_tokens	body_added_tokens	header_added_tokens	header_fraction	additional_passages
development	qwen7	409.46	257.00	152.47	0.37	6.82
development	llama8	363.75	244.54	119.21	0.33	6.82
heldout	qwen7	407.73	256.85	150.88	0.37	6.73
heldout	llama8	362.54	244.51	118.03	0.33	6.73

The header-cost explanation is plausible and quantitatively substantial. These measurements do not establish that headers caused particular answer errors. Under a shared full-prompt cap, more headers also leave fewer tokens for source content; the next experiments retain that constraint explicitly.

view	slots	unique_passages	duplicate_slots	linked_slots	linked_unique_passages	links_active	first_article_block_slots
development	20.00	13.03	6.97	8.86	5.17	0.95	8.71

view	slots	unique_passages	duplicate_slots	linked_slots	linked_unique_passages	links_active	first_article_block_slots
heldout	20.00	13.19	6.81	8.90	5.15	0.94	8.57

The 20 slots retain repeated corpus passages across search steps. They are slot-level candidates, not 20 unique documents. On held-out questions, about 94% activate the link rule; roughly 8.90 slots but only 5.15 distinct passages are linked on average. Deduplication is not part of the evaluated operator.

Priority allocation and prompt identity

reader	Tier 0 count	Tier 1 count	Tier 2 count	Tier 0 tokens	Tier 1 tokens	Tier 2 tokens
llama8	6.45	4.22	1.65	147.30	73.11	23.08
qwen7	6.45	4.22	1.65	155.46	76.15	24.22

Tier token lengths above are measured independently for each source segment and are descriptive; they need not sum to the actual full-prompt increment because token boundaries change when text is joined. The first-block identity audit finds that only about 30.7% of the added block-initial segments in the original completion come from the first block of their Wikipedia article. A block-initial segment is therefore not generally a document lead sentence.

reader	allowance	Identical pairs	Identical fraction	Questions
llama8	128	40	0.08	480
llama8	256	51	0.11	480
llama8	512	62	0.13	480
qwen7	128	38	0.08	480
qwen7	256	49	0.10	480
qwen7	512	61	0.13	480

Between 7.9% and 12.9% of the paired native-cap prompts are identical, depending on reader and allowance. Link activation by itself does not imply that the final selected context differs from proximity expansion.

reader	arm	allowance	Fact match (%)	Answer present (%)	Header tokens	Body tokens
llama8	native_linked	128	32.25	52.10	82.65	207.35
llama8	native_linked	256	34.23	54.36	118.77	299.39
llama8	native_linked	512	36.64	57.31	161.55	512.10
llama8	native_window	128	29.76	53.42	47.65	240.35
llama8	native_window	256	33.62	56.60	58.67	356.50
llama8	native_window	512	37.10	60.26	85.73	585.22
qwen7	native_linked	128	31.59	51.63	98.88	209.29
qwen7	native_linked	256	34.30	53.90	141.19	295.46
qwen7	native_linked	512	35.91	56.62	198.16	494.12
qwen7	native_window	128	29.62	53.42	59.93	245.67
qwen7	native_window	256	33.26	55.91	71.54	360.60
qwen7	native_window	512	36.78	59.58	104.62	583.53

At +128 and +256, title-linked completion generally preserves more literal support sentences than window, while window contains the non-Boolean answer string more often. At +512, window also has greater support-sentence coverage. These are descriptive mechanisms to test, not post hoc rules for choosing a budget.

Corpus and segment definition

The local corpus metadata identifies DPR psgs_w100: Wikipedia 20 December 2018, distributed as disjoint 100-word blocks. The source index has 21,015,324 blocks. The sentence splitter is an exact-span punctuation regular expression, with a trailing unpunctuated remainder retained. Source offsets are preserved, but a complete splitter segment can begin or end in the middle of a linguistic sentence because the corpus block is already truncated. Future manuscript wording should use complete source segments unless true sentence boundaries are separately established. The matched diagnostic applied character trimming to 9,249 of 15,588 prompts, as the review states; 6,881 final prompts still contain partial added segments, while other trimming operations removed complete extra segments. The original pipeline exposes at most 192 Qwen tokens per visible excerpt; the completion selector operates on the corresponding full raw blocks. The new baseline matrix includes both representations.

See the official DPR dataset card for the dump and block definition. The manuscript needs this corpus citation and explicit splitting, excerpt and duplicate-slot rules.

Design sensitivity and interaction

Setting	Reader	SE (pp)	80% MDE (pp)
Held-out matched	qwen7	0.63	1.76
Held-out matched	llama8	0.65	1.83
Development +128	qwen7	1.49	4.18
Development +256	qwen7	1.56	4.37
Development +512	qwen7	1.49	4.18
Development +128	llama8	1.52	4.26
Development +256	llama8	1.66	4.66
Development +512	llama8	1.57	4.39

These are normal-approximation design sensitivities at two-sided alpha 0.05 and 80% power, using the observed paired variances. They are not retrospective proof of an absent effect. The review's approximate sensitivity and HotpotQA variance argument are supported. For a single matched-reader contrast, estimated balanced sample requirements are 4,664/5,046 questions per dataset for a 1pp effect (Qwen/Llama), and 7,287/7,884 for 0.8pp. These values do not automatically apply to the proposed average over six correlated conditions.

Averaging all six native-cap EM differences gives 0.660pp with a question-clustered stratified bootstrap 95% interval [-1.076, 2.431]. The 10,000 bootstrap resamples preserve all six outcomes from each question; they are not six independent samples. The corresponding two-dataset plug-in requirement is 1515 questions per dataset for a 1pp average effect, with uncertainty in the pilot variance and no guarantee for MuSiQue.

Hypothesis	Wald statistic	df	p	Holm p
All six effects equal	16.046356	5	0.006713	0.006713
Reader by budget interaction	11.986274	2	0.002496	0.004992

The two Wald diagnostics were specified together for this review analysis and adjusted as a two-test family. They provide exploratory evidence of heterogeneity and a reader-by-budget interaction. Thus it would be too strong to dismiss all sign reversals as undifferentiated noise. Interaction evidence is separate from a positive average effect; it neither validates the best-looking cells nor confirms an adaptive budget policy. The existing six EM tests and confidence intervals remain unchanged.

Illustrative success and failure

The two examples below are selected by the first sorted question identifier among corrections and new errors for Qwen at the middle allowance (+256). They illustrate the observed behavior and are not a representative error taxonomy.

EM correction. Question: Where did Beatrice Of England's father die? Gold answer: Westminster. Window answer: Westminster, England. Completion answer: Westminster. Here the window answer already contains the gold location: this is an exact-match gain from answer specificity, not evidence that a missing fact was recovered.

New error. Question: Which film has the director died first, Choice Of Arms or Non Perdiamo La Testa? Gold answer: Non perdiamo la testa; Non Perdiamo La Testa. Window answer: Non Perdiamo La Testa. Completion answer: Choice Of Arms.

Follow-up and review corrections

The next development experiment contains 26,880 logical outputs: two readers, all three allowances, the two original native-cap operators, five priority ablations and LLMLingua-2, plus fresh seed, closed-book, visible-excerpt and full-source controls. The five ablations remove the link requirement, remove the first-segment tier, use only question links, use only seed links, or use one fixed random source ordering. All seed-retaining variants fit the full native prompt cap. No favorable budget is selected from the current results. This study remains exploratory.

A separate question-pool audit finds no unused HotpotQA validation questions, 90,434 unused HotpotQA train questions, 4,756 unused normalized 2Wiki dev questions and 1,457 unused MuSiQue dev questions. A new confirmation must state its actual source splits and cannot relabel the old held-out outcomes as fresh. The author has been asked whether to include an internally held-out sample from HotpotQA train.

Two formal criticisms require qualification. The current official DER-RAG entry explicitly lists DOI `10.18653/2026.findings-emnlp.1068`; the absence of `v1/` is not by itself an error. The Applied Intelligence instructions require a data availability statement and encourage a public repository; they do not universally mandate creating a public repository. The existing blank statements still need completion before submission.

The manuscript also needs standardized compound terms and terminology, citations for models/retrievers/inference software and adjacent methods, a source-selection overview, actual prompts and settings in place of operational paths, and a shorter discussion. These editorial repairs should accompany the completed evidence rather than make the current exploratory gains sound confirmed. No external submission or public release has occurred.